OThink-R1: Intrinsic Fast/Slow Thinking Mode Switching for Over-Reasoning Mitigation

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Abstract

Recent advanced large reasoning models (LRMs) leverage extended chain-of-thought (CoT) reasoning to solve complex tasks, achieving state-of-the-art performance. Despite their success, we identify a critical issue: a substantial portion of simple tasks solved by LRMs can also be addressed by non-reasoning LLMs using significantly fewer tokens, indicating the complex reasoning may not always be necessary. To address this, we systematically analyze the reasoning trajectories of LRMs and present a method utilizing identified paradigms and LLM-Judge to classify these trajectories as either Redundant Reasoning or Essential Reasoning. And we introduce OThink-R1, a method that prunes redundant reasoning steps while preserving logical validity. OThink-R1 dynamically employs the non-thinking mode (fast-thinking) for straightforward problems while engaging in deliberate thinking (slow-thinking) for complex problems. Experiments across mathematical and question-answering tasks demonstrate that OThink-R1 reduces reasoning redundancy by almost 23% on average without compromising accuracy, offering practical guidelines for efficient reasoning models. The code is available at https://github.com/AgenticIR-Lab/OThink-R1.

1 Introduction

Human cognitive processes are often divided into two complementary modes: fast, intuitive thinking (System 1) and slow, deliberate reasoning (System 2) [1]. Fast thinking is instinctive and effortless, allowing for quick decisions but often leading to cognitive biases in complex situations. In contrast, slow thinking is thoughtful and analytical, employing logical reasoning to achieve more accurate problem-solving. Traditional large language models (LLMs) primarily mimic fast thinking, relying on heuristics from vast data patterns, which can limit their effectiveness in handling intricate tasks. Recent advancements in language reasoning models (LRMs), such as OpenAI o1 [2], OpenAI o3-mini, and DeepSeek-R1 [3], emulate slow thinking by generating explicit reasoning chains (CoT) [4], transforming raw predictions into more structured, step-by-step solutions. This approach helps to mitigate the biases of fast thinking, significantly enhancing their reasoning capabilities for tackling complex problems.

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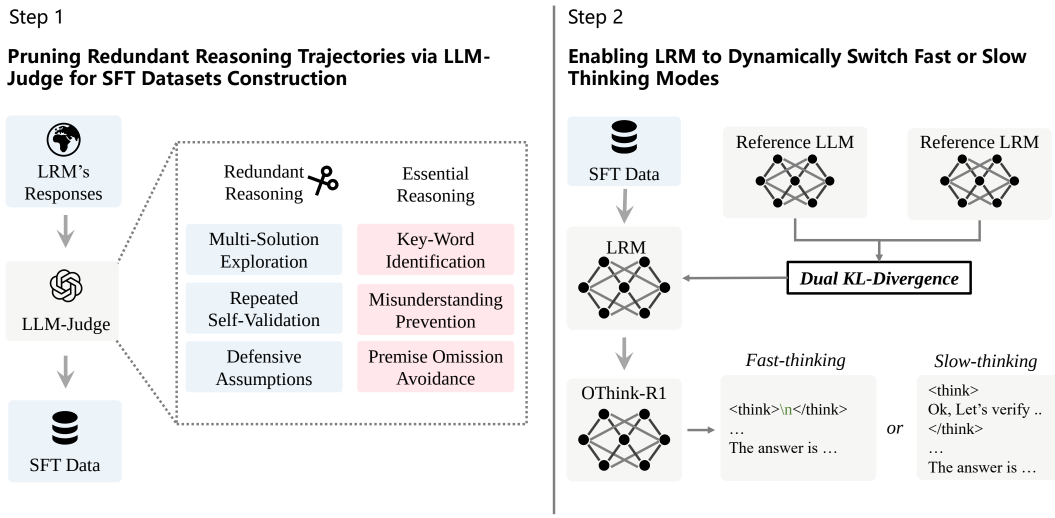

Figure 1: Schematic diagram of our proposed OThink-R1.

Despite the impressive performance of LRMs in complex tasks, they inevitably generate longer output sequences, which results in significantly higher computational overhead. For example, when comparing the typical LRMs (e.g. DeepSeek-R1-Distill-Qwen series) to the non-reasoning LLMs (e.g. Qwen2.5-Instruct series), we observe that the LRMs produce over 9.78 times more tokens than non-reasoning LLMs on both question answering (QA) and math tasks for the same correctly answered questions (Table 1). These substantial discrepancies in computational costs pose significant challenges to the widespread adoption of LRMs, raising concerns about inference latency and economic efficiency. Recent efforts have aimed to mitigate these issues through reinforcement learning techniques that control reasoning length [5–7] or prompts that eliminate reasoning steps [8]. However, these approaches confine models to single static reasoning patterns, either consistently compressed or entirely bypassed. This inflexibility prevents adaptive reasoning, where straightforward tasks should receive immediate responses while complex problems require thorough analysis, creating an intrinsic limitation in balancing computational efficiency with reasoning completeness.

To address this issue, we advocate for a new hybrid paradigm that integrates both fast-thinking and slow-thinking modes within a single model, enabling autonomous selection of the appropriate mode for different problems. This concept is inspired by cognitive theories [9], which postulates that humans possess both fast and slow thinking capacities and can flexibly switch between them as needed. Typically, fast thinking is used for a multitude of simple tasks, providing responses swiftly; while slow thinking is reserved for complex problems that demand detailed logical reasoning. We seek to transfer this paradigm to LRMs, allowing them to dynamically adapt their reasoning strategies. Our empirical results (Table 1) also indicate that slow thinking is not always necessary, as vanilla non-reasoning LLMs can solve a substantial portion of tasks with significantly fewer tokens. However, implementing adaptive hybrid reasoning in LRMs presents two key challenges: *1) How to identify which scenarios that fast thinking is sufficient? 2) How to empower LRMs to autonomously select the appropriate reasoning mode?* While a few pioneering works, such as Dualformer [10] and Qwen3[3], have explored hybrid modes, their methods still require manual selection of the reasoning mode and lack automatic adaptation.

In this work, we propose a novel framework, OThink-R1, which enables LRMs to automatically switch between fast and slow thinking modes. As shown in Figure 1, we systematically analyze the reasoning trajectories of LRMs across various tasks and reveal that certain slow-thinking trajectories are redundant and can be pruned to address the first challenge. Specifically, we identify three characteristic patterns in which LRMs generate less-essential output after arriving at the correct solution. Conversely, we also identify three essential reasoning patterns that significantly contribute to correct answers. These salient features are then formulated as prompts to instruct another LLM (e.g., GPT-4o) to serve as a judge and adaptively classify reasoning as either redundant or essential.

[3]https://github.com/QwenLM/Qwen3

To further enhance the ability of LRMs for automatic hybrid reasoning, we explicitly construct a supervised fine-tuning (SFT) dataset. For tasks where fast-thinking LLMs yield correct answers and the reasoning trajectories of LRMs are judged redundant, we remove the redundant reasoning trajectories, retaining only the immediate responses. These curated problem-response pairs are then used to fine-tune LRMs, thereby promoting their adaptive use of fast or slow thinking depending on the problem. Additionally, we propose a novel loss function that incorporates a dual-reference KL-divergence constraint, leveraging guidance from both fast and slow thinking variants of the same model architecture. This strengthens the model's dynamic switching ability between the two modes while simultaneously learning the output distributions of both reference models.

In our experiments, we implement our OThink-R1 into the representative LRMs, Deepseek-R1-Distill-Qwen series, with varying model sizes. Results show that OThink-R1 reduces generated tokens by an average of 23.4% on four representative QA and math datasets, while maintaining or even surpassing the performance of the original baselines. Further analysis reveals that the enhanced model employs fast thinking on more than 27.3% of problems. Our contributions are as follows:

- We propose OThink-R1, a novel large reasoning model that automatically switches between fast and slow thinking modes. This addresses a significant gap in the dynamic reasoning capabilities of existing models.
- We systematically analyze the reasoning trajectories of LRMs and present a method utilizing identified paradigms and LLM-Judge to classify these trajectories as either Redundant Reasoning (Multi-Solution Exploration, Repeated Self-Validation, Defensive Assumptions) or Essential Reasoning (Key-word Identification, Misunderstanding Prevention, Premise Omission Avoidance).
- We propose a dual reference KL-divergence loss function L_{hybrid}, incorporating guidance from fast and slow thinking models during supervised fine-tuning for LRMs. L_{hybrid} enables the integration of knowledge from multiple reference models into a unified reasoning model.
- We validated the effectiveness of the proposed OThink-R1 through systematic experiments on four public datasets. The experimental results demonstrate consistent improvements in performance with reduced token usage compared to the latest relevant baselines.

2 Preliminaries

Supervised Fine-Tuning. Let θ be the parameters of an LLM that predicts output sequences $\mathbf{y} = [y_1, ..., y_m]$ given input prompts $\mathbf{x} = [x_1, ..., x_n]$. SFT aims to refine the output style of pretrained LLMs, enabling them to generate target responses. Specifically, the pretrained model π_θ is finetuned on a curated dataset $\mathcal{D}_{\text{SFT}} = \{\mathbf{x}_{(i)}, \mathbf{y}_{(i)}\}_{i=1}^N$, via *maximum likelihood estimation* [11]:

$$\min_\theta \mathcal{L}(\theta, \mathcal{D}_{\text{SFT}}) = -\mathbb{E}_{(\mathbf{x},\mathbf{y})\in\mathcal{D}_{\text{SFT}}}\Big[\log \pi_\theta(\mathbf{y}|\mathbf{x})\Big] \quad (1)$$

where $\pi_\theta(\mathbf{y}|\mathbf{x})$ denotes the likelihood of the generated text, typically modeled auto-regressively as:

$$\pi_\theta(\mathbf{y}|\mathbf{x}) = \prod_{j=1}^{m} \pi_\theta(y_j|\mathbf{x}, \mathbf{y}_{<j}). \quad (2)$$

Thinking Mode. Cognitive processes in humans typically follow two modes: fast, intuitive thinking and slow, deliberate reasoning. This duality informs our understanding of the thinking modes employed by LRMs.

Slow-Thinking: This mode represents the traditional generation process, where models such as Deepseek-R1 and the Deepseek-R1-Distill-Qwen series engage in thorough analytical reasoning, leveraging structured logical processes. The slow-thinking mode consists of three key components: 1) a detailed reasoning process encapsulated within the <think> and </think> tags; 2) a step-by-step presentation of intermediate solutions; and 3) the final answer.

Fast-Thinking: In contrast to slow thinking, this mode emulates human instinctive cognition. For LRMs, the fast-thinking mode involves rapidly generating intermediate solutions and final answers while bypassing explicit reasoning steps, characterized by the empty content within the <think> tag.

Adaptive Hybrid-Thinking: To fully capture human cognitive flexibility, we introduce an Adaptive Hybrid-Thinking mode for LRMs. This mode empowers LRMs to autonomously switch between fast and slow thinking modes in response to task demands, optimizing performance based on task complexity.

Prior research has investigated hybrid approaches that integrate both fast and slow thinking modes. However, existing implementations generally require manual mode selection and show constrained adaptive behavior, which excludes them from the *Adaptive Hybrid-Thinking* mode.

3 Methodology

To enable LRMs to utilize an Adaptive Hybrid-Thinking mode, we propose OThink-R1, a model that dynamically switches between fast and slow thinking modes through the following three steps:

3.1 Identifying Paradigms of Complex Reasoning in LRMs

While LRMs demonstrate impressive reasoning abilities, our empirical experiments in Table 1 reveal that both non-reasoning LLMs (*e.g.*, Qwen2.5 series models) and LRMs can accurately solve a substantial portion of simple tasks (30% - 90% overlap in correct solutions), with non-reasoning LLMs requiring significantly fewer tokens. Crucially, this overlap motivates a deeper investigation: *Do the complex reasoning steps of LRMs matter when simpler non-reasoning LLMs can effectively solve the same tasks?* To address this, we systematically analyze the reasoning trajectories of LRMs on overlapping problems, and identify two distinct reasoning paradigms:

Redundant Reasoning: We observe three characteristic patterns of redundant reasoning (Appendix A) where LRMs produce non-essential output after determining a correct solution:

1. Multi-Solution Exploration: Persistent exploration of alternative solutions despite determining a correct answer.
2. Repeated Self-Validation: Excessive re-validation of every intermediate steps in final solutions.
3. Defensive Assumptions: Being overly cautious, take extraneous hypotheses into consideration based on internal knowledge, rather than problem-specific constraints.

Essential Reasoning: Building upon the pitfalls of redundant reasoning, we propose three fundamental principles for effective reasoning in LRMs based on observations (Appendix B). These principles address the limitations identified in redundant reasoning by emphasizing that reasoning steps should: 1) highlight critical problem elements; 2) resolve potential ambiguities; 3) strictly adhere to given constraints. Specifically:

1. Key-word Identification: Extracting and emphasizing critical problem components forms the basis for solution derivation.
2. Misunderstanding Prevention: Eliminating misunderstandings in the problem statement prevents errors from incorrect assumptions.
3. Premise Omission Avoidance: Comprehensive coverage of all given premises and conditions ensures solution validity.

3.2 SFT Dataset Construction with Curated Trajectories

As analyzed in Section 3.1, our observations indicate that reasoning is not always essential for solving certain problems. Both LRMs and non-reasoning LLMs can provide correct answers, with the latter doing so using significantly fewer tokens. Building on these findings, we focus on refining the training process by pruning redundant reasoning trajectories. The goal is to construct an SFT dataset that not only solves problems effectively but also enhances the model's ability to balance fast and slow thinking.

The construction of the SFT dataset begins with N query questions $\mathcal{Q} = \{\mathbf{x}_{(i)}\}_{i=1}^{N}$ and their corresponding responses $\mathcal{R} = \{\mathbf{y}_{(i)}\}_{i=1}^{N}$ generated by an LRM π_θ. Since the generated responses may inherently contain incorrect examples, directly incorporating them into training data risks

Table 1: Comparison of average generated tokens and overlap ratios between R1 (DeepSeek-R1-Distill-Qwen) and Qwen (Qwen2.5-Instruct). 1) **Qwen**: the average generated tokens of Qwen2.5-Instruct series. 2) **R1**: the average generated tokens of DeepSeek-R1-Distill-Qwen series. 2) **Overlap**: Percentage of problems that both R1 and Qwen can solve in training data (*a*/*b*), where *a* denotes the number of problems that both R1 and Qwen can solve, *b* denotes the number of training samples.

Dataset	1.5B			7B			14B		
	Qwen	R1	Overlap	Qwen	R1	Overlap	Qwen	R1	Overlap
OpenBookQA	45	802	**34.98%** (1734/4957)	36	638	**60.92%** (3020/4957)	89	559	**73.01%** (3619/4957)
CommonsenseQA	36	638	**35.21%** (2744/7793)	50	741	**51.20%** (3990/7793)	65	572	**61.62%** (4802/7793)
ASDIV	100	309	**86.04%** (826/960)	104	311	**90.10%** (865/960)	104	316	**88.75%** (852/960)
GSM8K	261	1025	**71.27%** (4261/5979)	276	691	**87.36%** (5223/5979)	274	649	**89.31%** (5340/5979)

reinforcing flawed reasoning patterns, which could derail the model's optimization trajectory. To ensure dataset quality, we filter the responses using a verification function $\mathtt{V}(\cdot)$, *e.g.*, Math-Verify[4], retaining only question-response pairs $(\mathbf{x}_{(i)}, \mathbf{y}_{(i)})$ that are verified as correct:

$$\mathcal{D} = \{(\mathbf{x}_{(i)}, \mathbf{y}_{(i)}) \mid \mathbf{x}_{(i)} \in \mathcal{Q},\ \mathbf{y}_{(i)} \in \mathcal{R},\ \mathtt{V}(\mathbf{y}_{(i)})\ \mathtt{is\ True}\}. \tag{3}$$

Next, we use a non-reasoning LLM of the same architecture to categorize the responses in $\mathcal{D}$ into two groups: (1) cases where both the LRM and the LLM provide correct answers, and (2) cases where only the LRM produces correct answers while the LLM fails.

For the first group, we employ an auxiliary verifier (LLM-Judge, implemented with GPT-4o) to identify reasoning paradigms. The LLM-Judge evaluates reasoning trajectories based on two criteria: (1) the presence of redundant reasoning patterns and (2) adherence to predefined essential reasoning principles. Detailed system prompt for identifying redundant or essential reasoning is provided in Appendix C.

If an LRM response is identified as exhibiting redundant reasoning, its reasoning trajectory, wrapped in `<think>`, is pruned during the construction of the SFT dataset and included as a fast-thinking sample. Otherwise, it is retained as a slow-thinking sample. Specifically, when constructing fast-thinking samples, we retain the `<think>` tags in the responses. This is because we have observed that the outputs of LRMs are highly dependent on `<think>` tokens (even if these tokens are removed from the chat template). Directly eliminating the `<think>` related tags can lead to fine-tuning failures.

Notably, for responses in the second group, we retain their full reasoning trajectories during fine-tuning. This preserves the model's capacity for complex reasoning while ensuring a comprehensive SFT dataset. The final SFT dataset combines pruned fast-thinking samples with unmodified slow-thinking trajectories, achieving an optimal balance between efficiency and robust reasoning ability.

3.3 SFT with Dual Reference KL-Divergence Loss

We have constructed the SFT dataset to guide LRMs in balancing their responses between fast and slow thinking modes. Traditional SFT that relies solely on maximum likelihood estimation often results in a preference for one predominant mode within the dataset, such as generating only fast or slow thinking responses. By explicitly leveraging the dual-mode characteristics within the dataset, we designed a specific loss function that enables the model to adopt the cautious reasoning style of LRMs during slow thinking while retaining the efficient generation capabilities of base LLMs during fast thinking.

Our approach encompasses two primary goals: (1) preventing excessive deviation from the original distribution of the base model during fine-tuning, and (2) ensuring that the model can dynamically adapt to the probability distributions of different thinking modes. To achieve this, we propose a dual-reference KL-divergence loss function:

$$\mathcal{L}_{\text{hybrid}} = -\mathbb{E}_{(\mathbf{x},\mathbf{y})\in\mathcal{D}_{\text{SFT}}}\Big[\log \pi_\theta(\mathbf{y}|\mathbf{x})\Big] + \beta_1 \text{KL}(\pi_\theta \| \pi_{\text{LRM}}) + \beta_2 \text{KL}(\pi_\theta \| \pi_{\text{LLM}}) \tag{4}$$

[4]https://github.com/huggingface/Math-Verify

The hyperparameters β_1 and β_2 regulate the retention level of the slow thinking mode relative to the reference distribution of LRMs, and the alignment strength of the fast thinking mode with the distribution of non-reasoning LLMs, respectively. Through the joint optimization of the supervised learning objective and the dual regularization terms, OThink-R1 gains the ability to autonomously select the optimal response mode based on context, thereby facilitating a dynamic balance between fast and slow thinking.

4 Experiments

4.1 Experimental Setting

OThink-R1's mode-switching efficacy was evaluated on simple tasks where: (1) fast thinking maintains performance with reduced token usage (unlike difficult tasks where it causes performance drops), and (2) the switching benefits are most demonstrable.

Datasets and Models. We evaluated our approach using relatively simple datasets from two tasks: question answering with OpenBookQA [12] and CommonsenseQA [13], and mathematical reasoning with ASDIV [14] and GSM8K [15]. Detailed descriptions of these datasets are provided in the Appendix D. The DeepSeek-R1-Distill-Qwen series (1.5B/7B/14B variants) served as foundational LRMs in our comparative experiments.

SFT Datasets Construction. We constructed two SFT datasets for training: a QA dataset combining OpenBookQA and CommonsenseQA and a MATH dataset merging ASDIV and GSM8K examples, with sample pruning ratios fully specified in Table 3a. All original training instances were processed through DeepSeek-R1-Distill-Qwen ($\tau = 0.9$, *top-p* $= 0.95$) to generate responses. The validation sets comprised 20% held-out samples from each dataset except for OpenBookQA which has a predefined 500-sample validation set.

Hyperparameter Tuning. The models are trained for up to 4 epochs with learning rates $\texttt{lr} \in [\texttt{3e}^{\texttt{-5}}, \texttt{4e}^{\texttt{-5}}, \texttt{5e}^{\texttt{-5}}]$. We use the AdamW optimizer [16] with default hyperparameters, and the gradient accumulation step is set to 2. Training is conducted on 8 × 80GB NVIDIA A100 GPUs. The details of the optimal hyperparameters are provided in Appendix E.

Baselines. We compare two baselines: 1) **NoThinking** [8], which directly bypass the reasoning process with explicit prompting; 2) **DualFormer** [10], which randomly drop the intermediate reasoning steps.

4.2 Main Results

Table 2: Performance comparison across multiple datasets, showing that OThink-R1 reduces the average number of generated tokens by 23.4% on QA and math tasks. (↓: fewer tokens, better efficiency; ↑: higher accuracy, better performance). Results for OThink-R1 are highlighted in blue, with bolded values indicating superior accuracy.

Model	OpenBookQA		CommonsenseQA		ASDIV		GSM8K	
	Tokens ↓	ACC ↑	Tokens ↓	ACC ↑	Tokens ↓	ACC ↑	Tokens ↓	ACC ↑
DeepSeek-R1-Distill-Qwen-1.5B	642	50.00%	528	43.80%	348	95.02%	1111	76.12%
NoThinking-R1-1.5B [8]	180	37.80%	163	37.70%	179	79.10%	404	69.70%
DualFormer-R1-1.5B [10]	399	49.00%	377	45.20%	203	86.70%	558	55.20%
OThink-R1-1.5B	206	**51.80%**	166	**48.40%**	271	**96.00%**	602	72.00%
DeepSeek-R1-Distill-Qwen-7B	783	76.40%	730	64.70%	352	97.00%	719	86.10%
NoThinking-R1-7B [8]	130	56.60%	106	54.30%	138	88.00%	258	77.50%
DualFormer-R1-7B [10]	723	75.20%	701	66.70%	223	96.00%	460	81.30%
OThink-R1-7B	667	**76.80%**	634	**66.90%**	270	**98.00%**	488	**86.70%**
DeepSeek-R1-Distill-Qwen-14B	522	92.80%	569	81.70%	319	97.00%	657	91.20%
NoThinking-R1-14B [8]	296	87.80%	373	78.90%	197	94.40%	458	88.60%
DualFormer-R1-14B [10]	1688	91.40%	2003	79.80%	223	95.70%	482	86.40%
OThink-R1-14B	421	**93.40%**	435	**81.80%**	412	**98.00%**	791	89.80%

Superior Efficiency-Effectiveness Trade-off in QA Tasks. As shown in Table 2, the OThink-R1 models demonstrate significant advantages over baseline LRM models across various parameter scales (1.5B, 7B, and 14B). Specifically, average accuracy improves by 3.2%, 1.3%, and 0.35%, while the average number of generated tokens decreases substantially by 68.2%, 13.9%, and 21.4%, respectively. Further analysis in Table 3b indicates that the model effectively activates the fast-slow thinking switching mechanism during QA processes, with this mechanism having a more pronounced impact in small parameter models. Comparative experimental results show that although the NoThinking generates the fewest tokens, it suffers from significant degradation in accuracy. Additionally, the DualFormer approach fails to deliver the anticipated performance improvements. In contrast, OThink-R1 maintains superior generation efficiency while consistently outperforming other comparative methods in accuracy, demonstrating the advantages of its dynamic fast-slow thinking switching capability.

Improved Performance in Simple Mathematical Tasks. In reasoning-intensive mathematical tasks, OThink-R1 consistently outperforms baseline LRM models on the ASDIV dataset (Table 2), achieving the highest accuracy across all model scales. When tackling the more challenging GSM8K dataset, the model effectively activates the fast-slow thinking switching mechanism (Table 3), with the 7B model generating fewer tokens while achieving higher accuracy. Notably, OThink-R1 surpasses both comparative methods in accuracy, demonstrating its ability to maintain excellent performance while enhancing efficiency—an advantage that extends even to reasoning-intensive tasks.

Table 3: Prune/Fast-Thinking Ratios Across Model Scales and Datasets. a. **Prune Ratio (Training)**: Pruning ratio of redundant reasoning trajectories within <think> during training (*a*/*b*), where *a* denotes the number of training examples whose reasoning trajectory was pruned, *b* denotes the training examples that R1 derive the correct answer. b. **Fast-thinking Ratio (Test)**: Percentage of fast thinking in testing set, average 27.3%.

a. Prune Ratio (Training)

Dataset	Prune Ratio (Training)		
	1.5B (%)	**7B (%)**	**14B (%)**
OpenBookQA	69.63 (1685/2420)	75.18 (2845/3784)	67.36 (3021/4485)
CommonsenseQA	76.31 (2631/3448)	74.89 (3684/4919)	62.70 (3870/6172)
ASDIV	30.15 (278/922)	32.51 (303/932)	31.45 (295/938)
GSM8K	36.51 (1818/4980)	32.11 (1734/5400)	31.65 (1738/5492)

b. Fast-thinking Ratio (Test)

Dataset	Fast-thinking Ratio (Test)		
	1.5B (%)	**7B (%)**	**14B (%)**
OpenBookQA	80.00	6.40	36.80
CommonsenseQA	80.34	8.76	35.70
ASDIV	18.27	13.62	7.97
GSM8K	20.77	9.93	8.56

4.3 Ablation Study

In Table 4, we evaluate various variants of OThink-R1: 1) w/o prune, which fine-tunes LRMs on filtered correct question-response pairs, without pruning any reasoning trajectories. 2) w/o LLM-Judge, where the LLM-Judge process is removed. This variation brute force prune the reasoning trajectories in cases where both the LRM and the LLM provide correct answers. 3) w/o KL-constraint employs the vanilla SFT loss (*cf.* Eq.(1)), without introducing dual KL-divergence constraint. 4) $\beta_1 = 0$, where the KL-constraint on reference LRM is removed. 5) $\beta_2 = 0$, where the KL-constraint on reference LLM is removed.

We observe that OThink-R1 outperforms other variants in most cases. We highlight the following key findings: 1) The only difference between w/o prune variant and OThink-R1 lies in the former retains all reasoning trajectories, which allows it to learn the external knowledge in the dataset. OThink-R1 outperforms w/o prune variant except in OpenBookQA on 14B series, demonstrating that the performance improvement stems from dynamical fast/slow thinking switching rather than dataset knowledge. 2) Compared to w/o LLM-Judge variant, OThink-R1 retains reasoning trajectories that is determined as essential. The result demonstrate that our systematic analyze on reasoning trajectories is indeed, as blindly removing essential reasoning trajectories leads to a performance drop. 3) OThink-R1 outperforms w/o KL-constraint on most cases, necessitating the utilization of dual KL-constraint. Moreover, we conduct fine-grained analysis by only remove one reference model. If we set $\beta_1 = 0$, *i.e.*, remove LRM reference model constraint, the model suffers from severe

Table 4: Ablation study across multiple datasets. (↓: lower is better, ↑: higher is better; Our proposed OThink-R1 is highlighted in blue, with bolded results indicating superior accuracy.)

Model	OpenBookQA		CommonsenseQA		ASDIV		GSM8K	
	Tokens ↓	**ACC ↑**	**Tokens ↓**	**ACC ↑**	**Tokens ↓**	**ACC ↑**	**Tokens ↓**	**ACC ↑**
w/o prune	734	51.40%	621	46.40%	298	93.40%	991	75.30%
w/o LLM-Judge	177	52.40%	213	49.10%	192	94.70%	406	69.10%
w/o KL-constraint	151	51.60%	182	47.10%	264	94.00%	564	71.90%
$\beta_1 = 0$	1776	47.40%	1355	48.70%	939	89.40%	1506	69.40%
$\beta_2 = 0$	172	47.80%	161	47.60%	261	94.00%	621	70.10%
OThink-R1-1.5B	206	51.80%	166	48.40%	347	94.00%	757	71.40%
w/o prune	997	75.40%	775	66.10%	249	98.00%	432	86.40%
w/o LLM-Judge	997	69.40%	776	65.70%	171	97.00%	279	84.20%
w/o KL-constraint	328	68.80%	268	64.00%	292	97.30%	588	86.40%
$\beta_1 = 0$	4364	72.20%	4320	69.40%	2144	97.70%	2073	84.20%
$\beta_2 = 0$	649	76.40%	617	64.20%	267	97.30%	480	85.50%
OThink-R1-7B	667	**76.80%**	634	66.90%	270	**98.00%**	488	**86.70%**
w/o prune	1676	94.60%	1455	80.70%	305	96.70%	629	91.70%
w/o LLM-Judge	4353	93.20%	5038	81.50%	196	98.00%	300	88.70%
w/o KL-constraint	3731	94.80%	3001	81.60%	281	97.00%	338	91.10%
$\beta_1 = 0$	10599	89.00%	11516	80.80%	8566	89.70%	4807	82.70%
$\beta_2 = 0$	253	91.40%	409	81.20%	282	96.30%	475	88.30%
OThink-R1-14B	421	93.40%	435	**81.80%**	412	**98.00%**	791	89.80%

overthinking issues, significantly improve the average generated tokens. If we remove LLM reference model constraint, the performance drops. These results necessate the dual KL-divergence constraint.

4.4 Case Study

We present the case study in Figure 2. OThink-R1 utilize fast-thinking mode, and solve the problem correctly. While the reasoning model exhibits overthinking phenomenon, significantly improve the generated tokens, while even derive the incorrect answer. It demonstrate that OThink-R1 can dynamically switch between fast and slow thinking. Moreover, the reasoning process may not always be necessary, which would even influence the model performance.

5 Related work

5.1 Efficient Reasoning

Recent advances in efficient reasoning for LRMs have attracted growing attention. Current approaches can be broadly categorized into *training* and training-free strategies.

Training Strategies. A prominent direction involves reinforcement learning with explicit length-based rewards [6, 17–19]. For instance, DAST [20] fine-tunes LRMs via a length-preference dataset governed by an adaptive token-length budget. Similarly, L1 [5] enforces length constraints by adding "Think for N tokens" instructions in training samples before policy optimization. Other efforts including training LLMs to learn latent representations [21–25], explicitly instruct LLMs to generate fewer reasoning steps [26–30] (*e.g.*, setting a token budget in prompts), or dynamically route inputs based on task difficulty to control reasoning complexity [31, 32]. However, these approaches share a critical limitation: they employ fixed compression strategies that lack the capacity for dynamic switching between different reasoning modes.

Training-free Strategies. Several works achieve efficiency through inference-time adjustments. NoThinking [8] suppresses intermediate reasoning steps via specialized prompting, while Dynam-

Question: If someone was making breakfast, they'd probably put two slices of bread in the what?
(A) plastic bag; (B) pantry; (C) supermarket; **(D) toaster**; (E) prison

DeepSeek-R1-Distill-Qwen-7B Response

<think>
…
Option B is the pantry. … That makes sense because it's a common place for storing and accessing breakfast items.
…
Option D is the toaster. … But the question is about where they put the bread when making breakfast, not using it in the toaster. So the bread would go into the toaster for that specific use, but not where you store it.
...
I think I'm confident with that. So the answer should be (B) pantry.
</think>
The answer is (B) pantry.

Whole Token Number: 675

OThink-R1-7B Response

<think>\n**</think>**
The correct answer is:
(D) toaster

Explanation:
When someone is making breakfast, they typically toast bread using a toaster. The toaster is the device that provides the heat needed to toast the bread slices, making it the most appropriate choice among the given options.

Whole Token Number: 57

Figure 2: Comparison of a generated responses on CommonsenseQA.

icExit [33] employs a pattern detection (*e.g.*, hesitation markers like "Wait" or "Hmm") to terminate reasoning early. However, these works cannot adaptively choose between fast and slow thinking.

5.2 Variable-Length Reasoning

Recent work explores supervised fine-tuning (SFT) on variable-length reasoning data to balance efficiency and accuracy of LRMs. C3oT [34] compresses extended CoT trajectories via GPT-4 distillation, retaining critical steps while shortening reasoning length. CoT-Valve [35] dynamically adjusts reasoning steps by interpolating parameters from long-reasoning and non-reasoning models. DualFormal [10] randomly prunes reasoning trajectories during training to reduce step redundancy. The most similar work is Yu et al. [36], which distill reasoning models into non-reasoning ones by fine-tuning solely on final answers. In contrast, OThink-R1selectively prunes reasoning trajectories while enabling LRMs dynamic switching between fast and slow thinking. Other efforts including TokenSkip [37] and iterative self-compression [38] similarly prioritize step reduction via semantic importance estimation or self-distillation. While these methods enhance efficiency through shortened reasoning trajectories, they primarily focus on static compression and lack explicit control over thinking modes, which OThink-R1overcomes.

5.3 Overthinking Issue

Recent work has explored the phenomenon of overthinking in LRMs, where models engage in redundant and unnecessary reasoning steps. Cuadron et al. [39] formalize this issue in code generation, demonstrating how LRMs prioritize prolonged internal reasoning over essential environmental interactions. Subsequent studies classify distinct types of overthinking: Fu et al. [40] identify self-doubt behaviors such as Reassessment and Perspective Bridging, while Qu et al. [41] analyzes underlying causes of redundant reasoning, including Circular Reasoning and Inefficient Loopback. Notably, Yi and Wang [42] propose a taxonomy including Continued Exploration of Alternative Strategies and Model Collapse, highlighting trade-offs between reasoning depth and output stability. Another study by Yang et al. [33] evaluates confidence in intermediate steps by detecting uncertainty markers (*e.g.*,"Hmm", "Wait" and others), offering a dynamic termination criterion. While existing works focus on characterizing causes or categorizing types of overthinking, our work moves beyond the issue itself, identifying the principles that enable LRMs to reliably derive solutions.

6 Limitations

Although our proposed OThink-R1 endows large reasoning models adaptively switch between fast-thinking and slowing, while significantly reduce the average generated tokens, we utilize LLM-Judge for determining redundant reasoning trajectories. In the future, how to end-to-end identify the redundant reasoning, and design more specified criteria is a promising direction.

7 Conclusion

We propose OThink-R1, a novel large reasoning model that dynamically switches between fast and slow thinking modes, improving both efficiency and performance. By pruning redundant reasoning trajectories and introducing a dual-reference KL-divergence loss, OThink-R1 enhances hybrid reasoning capabilities. Experimental results demonstrate its effectiveness in advancing adaptive and efficient AI reasoning across diverse tasks. These results highlight the framework's potential for efficient and scalable AI reasoning, with future work exploring extensions to multimodal reasoning and broader model architectures to further advance adaptive AI systems.

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Appendix

A Case Study on Redundant Reasoning

In this section, we present three redundant reasoning cases as follows:

Repeated Self-Validation: As shown in Figure 3, the LRM double-checks its own reasoning steps. It does this by going back over its previous conclusions, using phrases like "I double-checked it multiple ways" and "let me recap one more time". While this repeated checking helps ensure answers stay consistent, it also uses extra time and computational resources. Furthermore, once the model has already derived the correct answer, these extra checks are unnecessary.

Question

Joy can read 8 pages of a book in 20 minutes. How many hours will it take her to read 120 pages?

DeepSeek-R1-Distill-Qwen-7B (Generated Reasoning Tokens: 1416)

<think>

"...I think I've verified it enough ways. So, it's 5 hours. **But just to recap, because sometimes it's easy to get confused with rates and time**..."

"...**I don't think I made any mistakes here. I double-checked it multiple ways,** which is helpful because sometimes when you do math..."

"...Just to make sure, **let me recap one more time**:..."

</think>

Figure 3: Repeated Self-Validation on GSM8K.

Defensive Assumptions: As shown in Figure 4, the LRM acts too cautiously by exploring multiple hypotheses. Specifically, it questions its initial understanding by expressing "maybe she is asking" or "if there's another interpretation". Since the problem directly asks, "How many ounces of tea does she need?"—a straightforward question requiring the total number, these additional hypotheses are unnecessary. It wastes additional computational resources, reducing the inference efficiency.

Question

Artemis is making tea for a party. She knows her mom drinks an 8-ounce cup of tea and uses one ounce of tea. She will use this same ratio for the party. The party has 12 people there and each of them wants a 6-ounce cup of tea. How many ounces of tea does she need?

DeepSeek-R1-Distill-Qwen-7B (Generated Reasoning Tokens: 1316)

<think>

"...Alternatively, **maybe she is asking** how much tea she needs to add, not how much total tea is made?..."

"...**Wait, no, the question is**, how many ounces of tea does she need? ..."

"...**But let me think if there's another interpretation.** Maybe the 8-ounce cup is the total tea, including the tea itself and water?..."

</think>

Figure 4: Defensive Assumptions on GSM8K.

Multi-Solution Exploration: As illustrated in Figure 5, the model continues to explore alternative solutions even after deriving the correct answer. This behavior is evidenced by recurring phrases such as 'another way' 'Alternatively' and 'Another approach'. Since the model has derived the correct answer, exploring other solutions is redundant and wasting computational resources.

Question

Tim has 30 less apples than Martha, and Harry has half as many apples as Tim. If Martha has 68 apples, how many apples does Harry have?

DeepSeek-R1-Distill-Qwen-7B (Generated Reasoning Tokens: 818)

<think>

> "...Martha has 68, Tim has 30 less, so 68 - 30 = 38. Harry has half of Tim's apples, so 38 divided by 2 is 19. Yeah, that seems correct ..."
>
> "...Is there **another way** to approach this problem? Maybe by using algebraic expressions. Let me set variables for each person's apples ..."
>
> "...**Alternatively**, maybe I can visualize this with a diagram or something ..."
>
> "...**Another way** to think about it is by using equations ..."

</think>

Figure 5: Multi-Solution Exploration on GSM8K.

B Case Study on Essential Reasoning

In this section, we present three essential reasoning examples as follows:

Premise Omission Avoidance: The problem contains critical premises, and omitting any of these may lead to incorrect conclusions. As demonstrated in Figure 6, the model systematically identifies and incorporates each premise during reasoning. Specifically, it explicitly catches: 1) the weight of each item, and 2) What Tony's washing items includes.

Question

Tony's dad is very strict about the washing machine and family members are only allowed to wash 50 total ounces of clothing at a time. Tony doesn't want to break the rules, so he weighs his clothes and finds that a pair of socks weighs 2 ounces, underwear weighs 4 ounces, a shirt weighs 5 ounces, shorts weigh 8 ounces, and pants weigh 10 ounces. Tony is washing a pair of pants, 2 shirts, a pair of shorts, and 3 pairs of socks. How many more pairs of underwear can he add to the wash and not break the rule?

DeepSeek-R1-Distill-Qwen-7B (Generated Reasoning Tokens: 214)

<think>

"... **A pair of pants** weighs 10 ounces, **each shirt** weighs 5 ounces, **shorts** weigh 8 ounces, and **each pair of socks** weighs 2 ounces ..."

"... Tony is washing **one pair of pants**, **two shirts**, **one pair of shorts**, and **three pairs of socks** ..."

</think>

Figure 6: Premise Omission Avoidance on GSM8K.

Key-word Identification: The model performs reasoning to identify and analyze critical keywords within the problem statement. In Figure 7, the model explicitly recognizes "5 less than 20" and "one barnyard owl makes 5 hoots per minute", which enables it to derive the correct solution.

Question

One barnyard owl makes 5 hoot sounds per minute. If 5 less than 20 hoots per minute are heard coming out of the barn, how many Barnyard owls would this noise be coming from?

DeepSeek-R1-Distill-Qwen-7B (Generated Reasoning Tokens: 154)

<think>

"... First, I need to determine the total number of hoots per minute that are being heard coming out of the barn. ..."

"... The problem states that this number is **'5 less than 20.'** ..."

"... Next, I know that **one barnyard owl makes 5 hoots per minute** ..."

</think>

Figure 7: Key-word Identification on GSM8K.

Misunderstanding Prevention: The model explicitly verifies task requirements to avoid misunderstanding. As shown in Figure 8, when asked to calculate temperature decrease, the model recognizes this, rather than simply report the final temperature. It specifically highlights the phrases "lose one-fourth of its initial temperature" and "find the decrease", demonstrating awareness that the task requires computing the temperature change rather than deriving the resultant Addison mountain's temperature.

Question

In one hour, Addison mountain's temperature will decrease to 3/4 of its temperature. If the current temperature of the mountain is 84 degrees, what will the temperature decrease by?

DeepSeek-R1-Distill-Qwen-7B (Generated Reasoning Tokens: 94)

<think>

> "...To determine the temperature decrease, I first recognize that the temperature decreases to three-fourths of its current value after one hour ..." "...This means the temperature will **lose one-fourth of its initial temperature** ..."
>
> "...Given that the current temperature is 84 degrees, I calculate one-fourth of this value to **find the decrease** ..."

</think>

Figure 8: Misunderstanding Prevention on GSM8K.

C System Prompt

OThink-R1 employs an auxiliary verifier (LLM-Judge, implemented using GPT-4o) to distinguish between redundant reasoning and essential reasoning. To systematically categorize reasoning paradigms, given our observed redundant reasoning patterns and established essential reasoning principles, we construct the following LLM-Judge prompt:

System Prompt

Please act as a classifier for the following task.
Task Overview: You will receive a question along with its correct answer. The model's thought process will be provided within <think></think> tags. Your job is to evaluate whether the model's thinking is necessary and classify it accordingly.
Criteria for Classification:
Redundant reasoning: When the model has already derived the correct answer but continues to think, classify as follows:

1) **Multi-Solution Exploration**: The model has already derived the correct answer. However, the model continues to explore alternative solutions, using phrases like "Is there any chance" or "Another Way."
2) **Repeated Self-Validation**: The model has already derived the correct answer. However, the model continues to check or validates its previous intermediate reasoning steps for several times.
3) **Defensive Assumptions**: The model has already derived the correct answer. However, the model is overly cautious, continuing to propose extraneous hypotheses to confirm the correctness of its answer, which is irrelevant to problem-solving.

Essential Thinking: The essential reasoning follows the following principles:

1) **Key-word Identification**: The model provides concise analysis that helps quickly identify the key words, core elements of the problem.
2) **Misunderstanding Prevention**: The model conducts reasoning which aims to eliminate misunderstandings within the problem statement, thereby preventing errors stemming from incorrect assumptions.
3) **Premise Omission Avoidance**: The model conducts reasoning which confirms all necessary premises of problems to ensure understanding completeness.

Response Format:

- Correct answer: (The correct answer of the question)
- Is the model's thinking essential or redundant? (yes/no)
- If essential, the reason is?
- If redundant, the reason is?
- Thinking category: <answer> (Redundant/Essential) </answer>
- Classification: <subset> class of the response </subset>

Please analyze the model's thought process based on these criteria.

Figure 9: System Prompt

D Experiment Datasets

We evaluated our approach using the following four datasets:

- OpenBookQA [12]: OpenBookQA provides the additional open-book knowledge for solving the multiple-choice questions. It consists of 4.96K training pairs with 500 validation and test examples. In this study, we intentionally exclude this open-book information during training and test, using only question-answer pairs. This design choice isolates the evaluation of OThink-R1, as incorporating external knowledge risks conflating performance gains between retrieved knowledge and the method itself.
- CommonsenseQA [13]: CommonsenseQA consists of multiple-choice questions designed to evaluate models' ability to apply diverse forms of commonsense knowledge in reasoning tasks. The dataset is split into 9.4K training, 1.22K validation and 1.14K test sets. As there is no answer key in the test set, the validation set is utilized to evaluate the performance of our proposed OThink-R1.
- ASDIV [14]: A dataset of elementary math word problems comprising 1,200 training and 301 test problems. The test set includes diverse problem types designed to assess basic arithmetic operations, such as aggregation, addition, and others.
- GSM8K [15]: GSM8K is a linguistically diverse dataset of 8.5K grade school math word problems—comprising 7.5K training and 1K test examples—that is designed to evaluate multi-step reasoning.

E Optimal Hyperparameters for Tuning

We use the open-source TRL library [5] to train our models. The learning rate is searched in the range $\text{lr} \in \{3\times10^{-5}, 4\times10^{-5}, 5\times10^{-5}\}$, $\beta_1, \beta_2 \in \{10^{-3}, 10^{-4}\}$, **gas** (gradient accumulation steps) is searched in the range $\{1, 4\}$, **batch** (batch size in each GPU device) is set as 2. The maximum padding length for SFT is set as 3000. For all experiments, we train for 4 epochs, saving a checkpoint after each epoch. Final reported performance corresponds to the model checkpoint (selected by epoch) and hyperparameters that achieve the highest validation accuracy. The optimal hyperparameters for each method, across all datasets and model scales (1.5B, 7B, 14B), are listed in Tables 6 and 7, following the order of hyperparameters specified in Table 5.

Table 5: Hyperparameters to be searched for each method. **gas** denotes the *gradient accumulation steps*; **batch** denotes the *batch size in each GPU device*.

Method	Hyperparameters
DualFormer	**lr**, **gas**, **batch**
OThink-R1	**lr**, **gas**, **batch**, β_1, β_2

Table 6: Optimal hyperparameters and best-performing epoch checkpoint (selected by validation accuracy) on QA.

Model Size	Method	Hyperparameters					Checkpoint Epoch
1.5B	DualFormer	4×10^{-5}	4	2			3
	OThink-R1	5×10^{-5}	4	2	10^{-4}	10^{-4}	4
7B	DualFormer	3×10^{-5}	4	2			2
	OThink-R1	3×10^{-5}	4	2	10^{-3}	10^{-3}	1
14B	DualFormer	3×10^{-5}	4	2			2
	OThink-R1	3×10^{-5}	4	2	10^{-4}	10^{-4}	4

Table 7: Optimal hyperparameters and best-performing epoch checkpoint (selected by validation accuracy) on MATH.

Model Size	Method	Hyperparameters					Checkpoint Epoch
1.5B	DualFormer	5×10^{-5}	4	2			4
	OThink-R1	4×10^{-5}	1	2	10^{-3}	10^{-4}	4
7B	DualFormer	4×10^{-5}	4	2			2
	OThink-R1	3×10^{-5}	1	2	10^{-3}	10^{-4}	2
14B	DualFormer	3×10^{-5}	4	2			4
	OThink-R1	3×10^{-5}	1	2	10^{-4}	10^{-4}	3

For inference, we utilize the setting $\tau = 0.9$, *top-p* $= 0.95$ for 1.5B/7B/14B models. As 1.5B series occasionally enter *sentence-level loops* [43], we additionally set the `repetition_penalty` as 1.1. The LLM inference is conducted via the open-source vLLM library [6].

[5] `https://github.com/huggingface/trl`
[6] `https://github.com/vllm-project/vllm`

F Societal Impacts

Modern large reasoning models like DeepSeek-R1 demonstrate exceptional performance in complex reasoning tasks. However, their reliance on hybrid reasoning paradigms (i.e., slow-thinking mechanisms) incurs significant computational overhead for simple tasks that non-reasoning models can solve with far fewer generated tokens. This inefficiency raises concerns about resource consumption, particularly given the growing societal reliance on AI systems.

Our proposed OThink-R1addresses this challenge by enabling dynamic switching between fast- and slow-thinking modes. This adaptive approach reduces unnecessary computational expenditure while preserving performance on complex tasks. By optimizing resource utilization, our method contributes to:

- **Environmental sustainability**: Lower energy consumption mitigates the carbon footprint of large-scale AI deployments.
- **Maintenance efficiency**: OThink-R1 improve the inference efficiency, which decreases human labor costs in monitoring and maintaining computational resources.